Table 4. Random parameters binary logit model with heterogeneity in means and variances of crash injury severity (serious injury/fatal vs. PDO/minor injury), single-motorcycle crashes, Thailand, 2016–2023 (two-year periods), adaptive fused temporal structure.

Variable description	Estimated parameter	*t*-statistic	Marginal effects	
			PDO / minor injury	Serious injury / fatal
Constant [serious injury / fatal]				
Constant [2016–2021]	0.038	0.38		
Constant [2022–2023]	–0.183	–1.33		
Variables producing the same parameter value across all periods [2016–2023]				
Crash type				
Ran off road to the left and hit an object (1 if yes, 0 otherwise) [2016–2023]	0.499	3.01	–0.0640	0.0640
Vehicle mounted a traffic island (1 if yes, 0 otherwise) [2016–2023]	0.743	1.54	–0.0621	0.0621
(Standard deviation of parameter distribution; 61.4% above zero)	2.574	2.10		
Ran off road to the right (1 if yes, 0 otherwise) [2016–2023]	–0.394	–2.52	0.0511	–0.0511
Roadway characteristics				
Raised or earth median (1 if yes, 0 otherwise) [2016–2023]	–0.227	–1.96	0.0301	–0.0301
(Standard deviation of parameter distribution; 40.5% above zero)	0.941	1.99		
Depressed median (1 if yes, 0 otherwise) [2016–2023]	0.542	4.62	–0.0695	0.0695
Barrier median (1 if yes, 0 otherwise) [2016–2023]	–0.398	–2.84	0.0519	–0.0519
Concrete pavement surface (1 if yes, 0 otherwise) [2016–2023]	–0.875	–2.67	0.0583	–0.0583
(Standard deviation of parameter distribution; 40.9% above zero)	3.817	3.39		
Environmental characteristics				
Wet road surface (1 if yes, 0 otherwise) [2016–2023]	0.687	2.97	–0.0868	0.0868
Dark, with street lighting (1 if yes, 0 otherwise) [2016–2023]	1.171	9.03	–0.1471	0.1471
Temporal characteristics				
Songkran festival period (1 if yes, 0 otherwise) [2016–2023]	–1.178	–5.38	0.1166	–0.1166
(Standard deviation of parameter distribution; 39.4% above zero)	4.383	4.18		
New Year festival period (1 if yes, 0 otherwise) [2016–2023]	–1.132	–6.66	0.1229	–0.1229
(Standard deviation of parameter distribution; 34.4% above zero)	2.824	5.02		
Evening (1 if yes, 0 otherwise) [2016–2023]	–0.877	–7.01	0.1114	–0.1114
Variables with temporally varying parameters				
Crash type				
Ran off road to the left (1 if yes, 0 otherwise) [2018–2019]	–0.770	–4.62	0.0945	–0.0945
Ran off road to the left (1 if yes, 0 otherwise) [2020–2023]	–0.567	–4.49	0.0779	–0.0779
Loss of vehicle control (1 if yes, 0 otherwise) [2018–2023]	–1.616	–10.93	0.2143	–0.2143
Ran off road on a right bend (1 if yes, 0 otherwise) [2016–2017]	0.762	2.32	–0.0927	0.0927
Crash cause				
Speeding (1 if yes, 0 otherwise) [2018–2023]	0.295	2.97	–0.0389	0.0389
Cutting in too closely (1 if yes, 0 otherwise) [2018–2019]	0.803	2.25	–0.0944	0.0944
Cutting in too closely (1 if yes, 0 otherwise) [2020–2023]	–0.798	–3.05	0.1098	–0.1098
Driver fell asleep (1 if yes, 0 otherwise) [2022–2023]	1.909	3.23	–0.2293	0.2293
Drunk driving (1 if yes, 0 otherwise) [2016–2019]	–0.503	–2.39	0.0621	–0.0621
Roadway characteristics				
Crash on the main carriageway (1 if yes, 0 otherwise) [2016–2017]	–1.200	–3.53	0.1488	–0.1488
Crash on the main carriageway (1 if yes, 0 otherwise) [2018–2023]	0.621	2.50	–0.0798	0.0798
Crash on a frontage road (1 if yes, 0 otherwise) [2016–2019]	–1.679	–6.81	0.2031	–0.2031
Painted median (1 if yes, 0 otherwise) [2016–2019]	0.439	2.07	–0.0533	0.0533
Painted median (1 if yes, 0 otherwise) [2020–2023]	–0.352	–1.82	0.0485	–0.0485
Environmental characteristics				
Raining (1 if yes, 0 otherwise) [2020–2023]	–0.894	–2.83	0.1217	–0.1217
Dark, no street lighting (1 if yes, 0 otherwise) [2016–2017]	1.733	6.56	–0.2040	0.2040
Dark, no street lighting (1 if yes, 0 otherwise) [2018–2023]	1.246	7.91	–0.1569	0.1569
Region				
Eastern region (1 if yes, 0 otherwise) [2016–2021]	3.293	2.28	–0.1719	0.1719
(Standard deviation of parameter distribution; 71.2% above zero)	5.874	2.13		

Variable description	Estimated parameter	*t*-statistic	Marginal effects	
			PDO / minor injury	Serious injury / fatal
Northeastern region (1 if yes, 0 otherwise) [2016–2021]	1.342	5.00	–0.1277	0.1277
(Standard deviation of parameter distribution; 72.1% above zero)	2.292	4.02		
Northeastern region (1 if yes, 0 otherwise) [2022–2023]	1.428	6.46	–0.1895	0.1895
Western region (1 if yes, 0 otherwise) [2020–2023]	0.875	4.20	–0.1167	0.1167
Heterogeneity in the means of random parameters				
Concrete pavement surface [2016–2023]: Eastern region [2022–2023]	–3.280	–2.78		
Concrete pavement surface [2016–2023]: Midnight / early morning [2016–2023]	1.438	2.25		
Heterogeneity in the variances of random parameters				
Songkran festival period [2016–2023]: Northern region [2016–2023]	–1.074	–2.37		
Model statistics				
Number of observations		7,325		
Number of estimated parameters		45		
Log-likelihood at zero, LL(0)		–5077.30		
Log-likelihood with constant only, LL(C)		–5074.03		
Log-likelihood at convergence, LL(β)		–4534.14		
McFadden pseudo $\rho^2 = 1 - LL(\beta)/LL(0)$		0.107		
AIC / BIC		9,158.3 / 9,468.7		
Random parameter distribution; simulation		Normal; 2,000 Halton draws		

Notes. A positive parameter increases the probability of a serious injury/fatal outcome. Reference categories: Central region, daytime, no median, one motorcycle. Periods in brackets share one parameter value; for a variable with temporally varying parameters, periods not shown have a parameter of zero. Temporal structure selected by the adaptive fused lasso in the fixed-parameter binary logit (Stage 1): the sparsest penalty whose structure is not rejected at the 90% level against the separate-period logits ($LR < \chi^2_{0,90}(df)$). The stage at which the structure was chosen (fixed-parameter or random parameters model) is the one with the lower BIC among models not rejected at the 90% level against the fully unconstrained models (Table 3; all candidates in Table D1). Random parameters (at most eight), heterogeneity in means (at most three terms) and in variances (at most one term) were searched with 100 Halton draws at the 90% level and confirmed with 2,000 draws; non-significant parameters were then removed, so all parameters except constants and the means of random parameters are statistically significant at the 90% level (a parameter is kept random only while its standard deviation is significant). A variable used in a heterogeneity function enters only the random parameter shown, for crashes in the shared periods, and is removed from the main utility. Marginal effects are average changes in the simulated unconditional probability (2,000 Halton draws) when the indicator switches from 0 to 1, averaged over crashes in the parameter's periods; for random parameters they include the heterogeneity terms. PDO/minor = –(serious/fatal). Percentage above zero evaluated at the heterogeneity variables equal to zero.